



HWLab-work#2

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1- Task 1:

```
#define GPIO0DIR *((volatile unsigned long *)0x50008000)

#define GPIO0DATA *((volatile unsigned long *)0x50003FFC)

#define GPIO2DIR *((volatile unsigned long *)0x50028000)

#define GPIO2DATA *((volatile unsigned long *)0x50023FFC)
```

```
int dec_to_7seg[] = {
```

```
    0b0111111,
```

```
    0b0000110,
```

```
    0b1011011,
```

```
    0b1001111,
```

```
    0b1100110,
```

```
    0b1101101,
```

```
    0b1111101,
```

```
    0b0000111,
```

```
    0b1111111,
```

```
    0b1101111
```

```
};
```

```
void delay(void) {
```

```
    int i;
```

```
    for (i = 0; i < 10000; i++);
```

```
}
```

```
int main(void) {
```

```
    int i, j;
```

```
    GPIO0DIR = 0b110;
```

```
    GPIO2DIR = 0b1111111;
```

```
    while (1) {
```

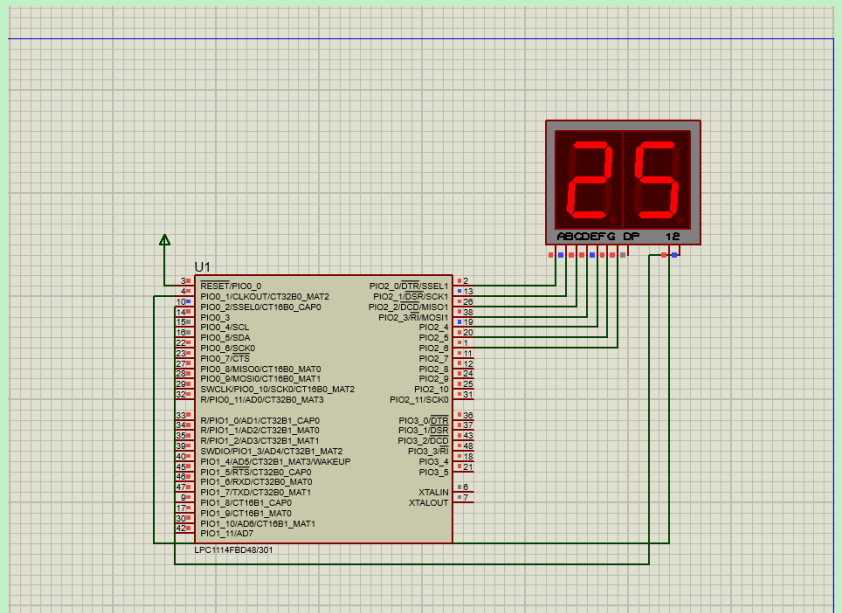
```
        GPIO0DATA = 0b010;
```

```
        GPIO2DATA = dec_to_7seg[5];
```

```
        for (i = 0; i < 100; i++);
```

```
        GPIO0DATA = 0b100;
```

```
        GPIO2DATA = dec_to_7seg[2];
```



```

        for (j = 0; j < 100; j++);

    }

}

```

2 – Task 2

```

#define GPIO0DIR *((volatile unsigned long *) 0x50008000)

#define GPIO0DATA *((volatile unsigned long *) 0x50003FFC)

#define GPIO1DIR *((volatile unsigned long *) 0x50018000)

#define GPIO1DATA *((volatile unsigned long *) 0x50013FFC)

#define GPIO2DIR *((volatile unsigned long *) 0x50028000)

#define GPIO2DATA *((volatile unsigned long *) 0x50023FFC)

```

```

int seven_seg_map[] = {

    0b1000000,

    0b1111001,

    0b0100100,

    0b0110000,

    0b0011001,

    0b0010010,

    0b0000010,

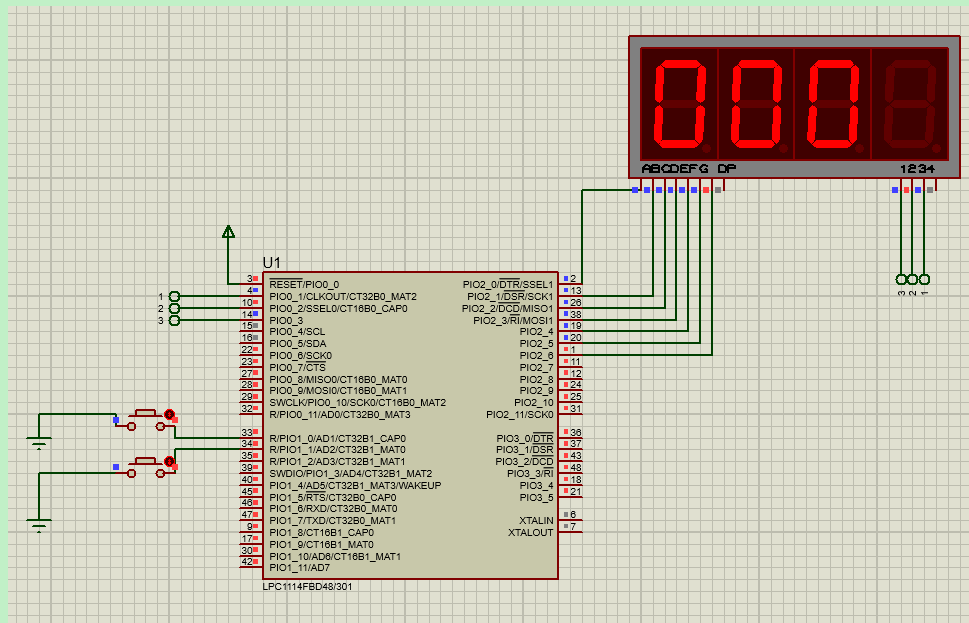
    0b1111000,

    0b0000000,

    0b0010000,

};

```



```

void show_number(int num) {

    int i;

    GPIO0DATA = 0b0010;

    GPIO2DATA = seven_seg_map[num % 10];

    for(i = 0; i < 100; i++);

    GPIO0DATA = 0b0100;

    GPIO2DATA = seven_seg_map[(num / 10) % 10];
}

```

```

for(i = 0; i < 100; i++);

GPIO0DATA = 0b1000;

GPIO2DATA = seven_seg_map[(num / 100) % 10];

for(i = 0; i < 100; i++);
}

```

```

int main(void) {

    int i;

    GPIO0DIR |= 0b1110;

    GPIO0DATA = 0b1110;

    GPIO2DIR |= 0b1111111;

    GPIO1DIR |= 0b11;

    int value = 0;

    int last_count = 0;

    while(1) {

        int inc = !(GPIO1DATA & 0b01);

        int rst = !(GPIO1DATA & 0b10);

        if(inc && !last_count) {

            value = (value + 1) % 1000;

        }

        if(rst) {

            value = 0;

        }

        last_count = inc;

        for(i = 0; i < 100; i++) {

            show_number(value);

        }

    }

}

```

